

QALHAT, Oman

WMO: 412670

Lat: 22.667N

Lon: 59.400E

Elev: 11

StdP: 101.19

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	19.2	1.3	4.2	28.5	3.8	5.0	27.1	20.5	23.0	16.7	22.6	2.8	280	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.3	40.7	23.1	39.1	23.5	37.9	23.8	30.1	33.7	29.5	33.4	29.0	33.1	5.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	26.0	32.7	28.5	24.9	32.4	27.9	23.9	32.3	100.9	33.7	97.8	33.5	95.2	33.2	35.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	9.8	8.1	16.9	44.4	0.8	2.0	16.3	45.8	15.8	47.0	15.4	48.1	14.8	49.5		
(5)			9.5	30.5	1.0	0.9	8.7	31.1	8.1	31.7	7.5	32.2	6.8	32.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	22.1	23.0	25.7	29.6	34.0	34.1	33.1	31.5	31.3	29.7	26.4	23.5
	DBStd	4.61	1.22	1.94	2.17	2.39	2.46	2.42	2.56	2.16	1.48	1.52	1.35	1.27
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	1	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6818	376	364	487	587	743	723	715	666	638	611	491	418
	CDD18.3	3776	118	131	229	337	485	473	457	407	388	352	241	160
	CDH23.3	48761	244	551	1928	4427	7852	7615	7045	5899	5578	4706	2241	676
	CDH26.7	27248	7	69	468	2152	5373	5216	4567	3425	3179	2274	488	29
Wind	WSAvg	3.1	4.0	4.1	3.4	2.9	2.9	2.9	2.7	2.5	2.3	2.5	3.4	4.1
Precipitation	PrecAvg	105	8	10	19	18	6	19	4	3	1	3	6	9
	PrecMax	272	87	92	108	54	45	223	38	15	4	27	77	83
	PrecMin	13	0	0	0	1	0	0	0	0	0	0	0	0
	PrecStd	67	17	19	27	17	11	47	9	3	1	7	17	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.5	31.1	35.0	37.9	42.2	42.6	43.3	40.5	37.8	35.7	32.1	28.6
		MCWB	19.2	17.7	19.1	20.0	22.6	22.6	23.2	23.9	22.5	21.2	19.7	19.6
	2%	DB	26.1	28.6	31.7	36.0	40.0	40.2	40.1	37.1	35.4	34.0	30.5	27.4
		MCWB	19.5	19.4	19.7	20.3	22.7	22.8	25.0	25.3	23.8	22.8	21.6	20.1
	5%	DB	25.4	27.0	30.2	34.4	38.7	38.7	37.7	35.4	34.3	33.0	29.7	26.6
		MCWB	19.4	19.9	19.9	21.1	22.8	23.9	25.5	25.7	24.7	23.7	22.0	20.1
	10%	DB	24.7	26.0	29.0	33.2	37.6	37.5	36.3	34.2	33.4	32.2	29.0	26.0
		MCWB	19.0	19.8	20.6	21.8	23.0	24.7	26.3	26.2	25.3	23.6	21.9	19.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	23.2	24.9	26.8	29.7	30.5	31.7	30.0	29.6	27.6	25.1	22.6
		MCDB	25.5	26.1	28.5	30.8	34.8	33.8	34.4	33.5	32.0	31.3	29.0	26.4
	2%	WB	21.0	22.4	24.0	26.1	28.7	29.7	30.3	29.2	28.7	26.8	24.3	21.8
		MCDB	24.8	25.7	27.7	30.4	34.0	33.9	33.9	32.7	32.0	31.2	28.7	25.9
	5%	WB	20.4	21.6	23.3	25.5	28.0	29.1	29.7	28.7	28.1	26.2	23.6	21.2
		MCDB	24.4	25.3	27.2	30.1	33.5	33.8	33.4	32.3	31.8	31.0	28.3	25.6
	10%	WB	19.7	20.9	22.7	24.9	27.3	28.6	29.1	28.2	27.6	25.6	23.0	20.6
		MCDB	24.0	24.9	26.9	29.9	33.4	33.6	33.1	31.9	31.6	30.7	27.9	25.3
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.7	4.9	4.9	5.3	4.8	4.2	4.0	3.9	4.6	4.5	4.3
		MCDBR	5.5	6.1	6.8	6.7	7.1	7.2	7.2	6.4	5.4	5.3	5.3	5.0
		MCWBR	4.4	5.3	6.2	6.5	6.3	6.6	6.6	5.9	5.5	6.2	7.1	6.3
	5% WB	MCDBR	4.7	4.6	4.6	4.9	5.4	4.7	4.4	4.2	3.8	4.2	4.5	4.5
		MCWBR	4.4	4.9	5.6	6.3	6.6	6.4	5.6	5.0	5.6	6.9	6.0	4.3
Clear-Sky Solar Irradiance	taub	0.445	0.493	0.563	0.632	0.680	0.764	0.835	0.775	0.649	0.509	0.469	0.443	
	taud	2.119	1.970	1.793	1.683	1.606	1.488	1.423	1.496	1.666	1.930	2.056	2.119	
	Ebn at Noon	809	793	756	713	676	616	574	609	681	767	776	792	
	Edh at Noon	143	176	220	249	268	299	319	297	246	181	151	138	
All-Sky Solar Radiation	RadAvg	4.46	5.27	6.08	6.68	7.15	6.74	6.23	6.24	6.08	5.58	4.67	4.24	
	RadStd	0.19	0.20	0.28	0.22	0.11	0.28	0.32	0.16	0.11	0.14	0.16	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.23	N/A	N/A	N/A	+0.39	+0.45	N/A	N/A	+96	+98

Nomenclature: See separate page